

Date	2 July 2026
Team ID	-
Project Name	Book-Store
Maximum Marks	4 Marks

PROJECT DESIGN PHASE-II: TECHNOLOGY STACK

Architecture & System Stack

The Book Store (BookVerse) application is engineered utilizing the MERN stack (MongoDB, Express, React, Node) to deliver a highly responsive, secure web interface. This architecture decouples frontend rendering from backend business logic controllers, facilitating database interactions via Mongoose ODM.

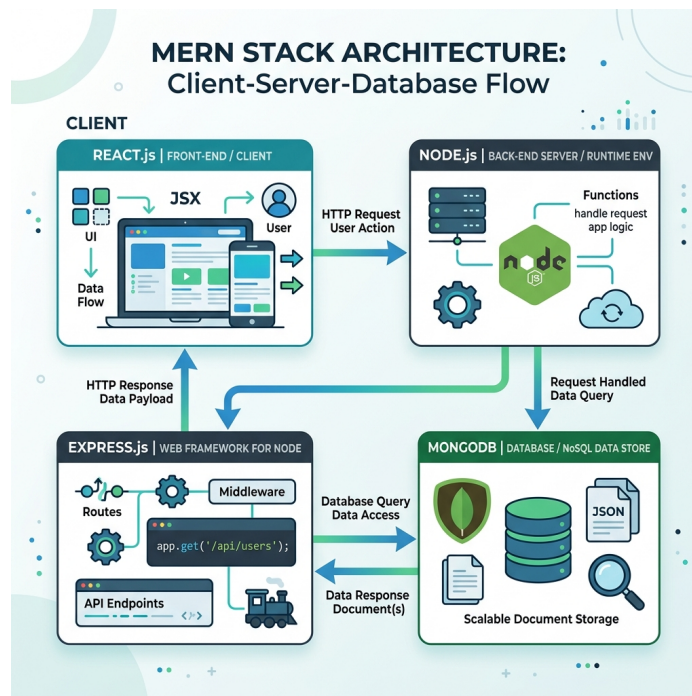


Figure 1: MERN Stack system architecture diagram illustrating data flows between clients and clusters.

Technology Stack Components

1. Frontend Layer (React & Vite)

Constructed using React.js bootstrapped with Vite. Global authentication contexts manage user login, signup, and credentials tokens. Viewports are styled with a warm cream (#FDFBF7) and brown (#7D4016) palette matching Bootstrap configurations.

2. Backend Layer (Node.js & Express.js)

Written as an Express server running on Node.js runtime. Business handlers are separated under MVC design patterns, and routes are protected using custom JWT token parsers and Multer file upload middlewares.

3. Database Storage Layer (MongoDB Atlas)

Connected to a cloud-hosted MongoDB Atlas cluster. Data models are designed with schema validations (Admins, Sellers, Users, Books, Orders), and queried response times are kept under 120ms.
